Ingrid Solberg

Financial Analyst — Quantitative Analysis

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PROFESSIONAL SUMMARY

Quantitative financial analyst with 12 years of experience in derivatives pricing, systematic portfolio research, and commodity forecasting. Builds models in Python and C++ using stochastic calculus, Monte Carlo simulation, time-series econometrics, and convex optimization, with documentation written for model validation. Developed a multi-factor equity risk model and a portfolio construction framework with transaction cost modeling at an asset manager, and reduced FX option calibration time from 40 to 6 seconds at a bank. Treats research with discipline: out-of-sample testing, realistic cost assumptions, and a clear distinction between simulated backtests and live performance. Works closely with portfolio managers, risk teams, and model validators, and is interested in volatility modeling and robust portfolio optimization.

ANALYTICAL AND TECHNICAL SKILLS

Quantitative methods: Stochastic calculus, Monte Carlo simulation, finite-difference methods, time-series econometrics (ARIMA, GARCH, regime-switching models), factor models, convex optimization, Bayesian estimation.

Derivatives and pricing: FX and equity options, local and stochastic volatility models, yield curve construction (Nelson-Siegel-Svensson), Greeks and hedging analysis.

Portfolio and risk analytics: Multi-factor risk models, portfolio optimization with transaction costs, performance attribution, Value at Risk and stress testing.

Programming and data analytics: Advanced: Python (NumPy, pandas, SciPy, statsmodels, cvxpy), C++17, SQL. Working knowledge: R, MATLAB, kdb+/q.

Financial modeling: Simulation-based scenario analysis, forward curve models for power and gas.

Tools and data: Bloomberg, Refinitiv, Git, Jupyter, pytest, Linux.

Model governance: Model documentation, independent validation support, backtesting standards.

Industry experience: Asset management, investment banking, energy commodity trading.

PROFESSIONAL EXPERIENCE

Limmat Asset Partners (asset management) — Zurich, Switzerland

Senior Quantitative Analyst — *Jun 2020 – Present*

- Developed a multi-factor equity risk model covering about 2,500 global stocks, with factor exposures, covariance shrinkage, and specific risk estimates; risk forecasts are used in daily portfolio reviews.
- Built a portfolio construction framework in Python with cvxpy that includes turnover limits and a transaction cost model calibrated on internal execution data, reducing estimated rebalancing costs by about 12 basis points per year in simulation.
- Designed a research backtesting standard with the risk team covering point-in-time data, out-of-sample periods, and cost assumptions; all research results are labeled as simulated and are not presented as live performance.
- Prepare model documentation and support the independent model validation team, resolving all findings from two annual reviews.
- Mentor two junior quantitative analysts on code review, testing, and research hygiene.

Helvetic Merchant Bank (investment banking) — Zurich, Switzerland

Quantitative Analyst, FX Derivatives — *Sep 2016 – May 2020*

- Implemented FX option pricing under local volatility with Monte Carlo and finite-difference methods in C++ with Python bindings, cutting calibration time from about 40 seconds to 6 seconds per currency pair.
- Built tools that explain daily profit and loss for the FX options desk by Greek and market move, reducing time spent investigating unexplained P&L for traders and product control.
- Supported credit valuation adjustment (CVA) calculations by providing exposure simulations for FX forwards and options.
- Wrote unit and regression tests for the pricing library, raising test coverage of the FX module from 45% to 80%.

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Nordfjord Commodities (energy trading) Oslo, Norway
Quantitative Research Analyst *Aug 2014 – Aug 2016*

- Built forward curve models for Nordic power and gas with seasonality and hydrological inputs, used by traders for daily valuations.
- Developed GARCH-based volatility forecasts for power prices used in risk limits.
- Automated daily data collection from exchanges and weather sources in Python.

SELECTED PROJECTS

Regime-switching volatility model
Tools and methods: Python, statsmodels, Markov-switching models, out-of-sample testing

- **Problem:** Single-regime volatility estimates reacted too slowly to market stress.
- **Contribution:** Implemented and tested a two-regime model for equity index volatility against a GARCH(1,1) baseline.
- **Result:** In out-of-sample tests, forecast error during stress periods fell by about 15% versus the baseline (research simulation, not trading results).

curvefit — yield curve library (open source)
Tools and methods: Python, NumPy, SciPy, pytest

- **Problem:** Teams used inconsistent spreadsheet implementations of the Nelson-Siegel-Svensson model.
- **Contribution:** Wrote a tested library with stable parameter estimation and documentation.
- **Result:** Adopted by the fixed income and risk teams at Limmat Asset Partners.

Execution cost study
Tools and methods: Python, SQL, market-impact modeling on internal execution data

- **Problem:** Transaction cost assumptions used in research were outdated.
- **Contribution:** Estimated a market-impact model from two years of internal trades and documented its limits.
- **Result:** The updated cost assumptions are now used in all portfolio research.

EDUCATION

Bergen School of Mathematical Sciences Bergen, Norway
M.Sc. Mathematical Finance *2012 – 2014*

- Thesis: calibration stability of stochastic volatility models for FX options.
- Coursework: stochastic calculus, numerical methods, time-series analysis, optimization.

Bergen School of Mathematical Sciences Bergen, Norway
B.Sc. Applied Mathematics *2009 – 2012*

- Coursework: probability theory, linear algebra, statistics, scientific computing.

HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **Quant Research Symposium Zurich (2024)** — Speaker on avoiding overfitting in factor research.
- **Algorithmic Trading Hackathon, Oslo (2019)** — Built a market-making strategy in a simulated exchange environment; results came from simulation only.
- **Women in Quantitative Finance Circle, Zurich (2021 – Present)** — Volunteer mentor for students and early-career analysts.

CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

Professional development (illustrative): Advanced Derivatives Modelling Course, Northbridge Finance Institute (2017); Machine Learning for Asset Management Program, Lakeshore Professional Institute (2022).

Open source: Maintainer of the curvefit yield curve library.

Languages: Norwegian (native), English (fluent), German (professional working proficiency).

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